

FULL-MANUSCRIPT QUALITY-ASSURANCE AUDIT

Locked Gravitational Saturation Cosmology

Audited file: Locked_Saturation_Cosmology_JournalReady_with_Copyright_Page.pdf

417 PDF pages | approximately 81,000 extracted words

SHA-256: e78c2c89438c64606c7bcf652c99db5e777062793526b355d4718beb45b82632

Audit scope. Full-document extraction and automated consistency scans were run across all 417 pages, followed by targeted inspection of the core derivations, locked parameter card, numerical result tables, end matter, appendices, and bibliography. The audit searched for duplicate passages, unresolved placeholders, parameter drift, numbering problems, visible editorial artifacts, internal arithmetic inconsistencies, and mathematical claim-boundary problems.

Bottom line. The manuscript is structurally coherent and the principal locked numerical values are internally consistent. No accidental duplicated long paragraphs, unresolved TODO markers, or obvious locked-parameter substitutions were found. However, several corrections should be made before university or journal circulation. Three are substantive mathematical-scope issues; the remainder are editorial, reproducibility, and presentation issues.

1. Findings requiring correction before submission

Severity	Location	Finding	Required correction
HIGH	Chs. 2, 5; App. F	The manuscript labels the displayed Einstein-plus-scalar expression as the complete gravitational action, but the saturated Friedmann factor is then imposed separately. The modified background constraint does not follow from the displayed action.	Call it the matter/scalar-sector action or an effective partial action. State explicitly that the saturated Friedmann constraint is an independent effective postulate pending a covariant completion.
HIGH	Bounce derivation; Chs. 2, 5, 6; Apps. A, F	The Raychaudhuri expression is obtained by differentiating $H^2=f(\rho)$. Algebraically deriving it normally divides by H , which is not valid exactly at $H=0$. The formula can be used at the bounce only by a stated continuous-limit argument or as an	Add a sentence proving the limiting extension to $H=0$, or define the Raychaudhuri equation independently as part of the effective system. Do not say the bounce value follows merely by differentiation at $H=0$.

		independently postulated evolution equation.	
HIGH	Stability claims; Chs. 13-14; Apps. D-E	The ghost/gradient-stability check covers the two canonical scalar fields. It does not establish stability of an unknown covariant gravitational completion responsible for the modified Friedmann equation.	Qualify every global stability claim: “the retained canonical scalar perturbation sector is stable on the expanding branch.” State that full gravitational-sector stability requires a covariant completion.
MEDIUM	Boltzmann and “complete/full” claims	The calculation is a full Boltzmann evolution of the locked expanding background and canonical fields, but it is not a perturbative evolution through the bounce derived from a complete covariant gravity theory.	Use “full Boltzmann evolution of the post-bounce locked background” and reserve “complete theory” for the effective model as defined, not a fundamental completion.
MEDIUM	Title page and front matter	The title page says “409+ pages,” while the original manuscript has 416 physical PDF pages and the copyright-added file has 417.	Replace with an exact version statement, e.g. “416-page technical manuscript; 417 pages in the copyright-notice edition.”
MEDIUM	Data and code statements	The manuscript says reproduction materials and solver outputs are specified, but the distributed PDF contains no code archive, parameter files, CAMB patch, output tables, or immutable repository identifier.	Before submission, publish the frozen code/data package and add a repository URL, release tag, checksum, software versions, and exact run commands.
MEDIUM	Throughout; especially list displays around PDF pp. 28, 113, 129, 148, 158, 178-179, 194, 239, 259-264, 333, 363-377	Many displayed prose lists contain a visible space before commas and periods, for example “the smooth early splice ,” and “the canonical saturation stress-energy .”	Remove spaces before punctuation in prose lists. This appears to originate from math/display formatting and is visible in the PDF.
LOW	Bibliography, final page	The Einstein title renders as “Relativit”	Correct the German title glyph, normalize

		atstheorie,” with a broken diacritic/spacing artifact. Several journal abbreviations also have spaces before commas. Planck lensing is entered twice by design.	journal punctuation, and cite the Planck lensing paper once unless the journal style explicitly permits repeated entries.
LOW	PDF navigation	The PDF has no document outline/bookmarks despite 14 chapters and six appendices.	Add PDF bookmarks for front matter, chapters, appendices, and bibliography. This is especially important for a 417-page review copy.
LOW	Copyright page	The added copyright page uses unembedded Helvetica while the manuscript fonts are embedded.	Embed the copyright-page fonts when rebuilding the final release PDF. This is not a content error but improves archival portability.

2. Checks that passed

- **Locked parameter consistency:** $C_s=0.95$, $V_{\infty}=2.187987$, $\rho_s=1.58199 \times 10^{15}$, $H_0=67.4$, $\Omega_{m0}=0.314114$, $\sigma_8=0.779680$, $S_8=0.79781$, $100 \theta^*=1.041033$, and $r_{\text{drag}}=149.960$ Mpc recur consistently. Apparent alternatives are explicitly labeled control-model values, sensitivity runs, or transition scans.
- **Numerical arithmetic:** S_8 recomputes to 0.7978099; the drag-horizon increase recomputes to 1.6657%; the locked/control S_8 reduction recomputes to 1.8729%; and the acoustic-reference offset recomputes to -0.00644%. These agree with the rounded manuscript values.
- **Energy-transfer bookkeeping:** The field loss term $-Q$ and radiation gain term $+Q$ cancel in the summed continuity equation. Matter is separately conserved in the stated system.
- **Low-density limit:** The saturated Friedmann factor tends to one for ρ/ρ_s much less than one, recovering the stated standard Friedmann limit.
- **Potential minimum:** At $C=C_s$ and $\phi=0$, the first derivatives vanish and the stated positive curvatures give a local minimum when $m_C^2 > 0$ and $A_0 > 0$.
- **Duplicate-text scan:** No accidental duplicated long paragraphs were detected. Five repeated full sentences are summary statements reused in different sections, not copy-paste blocks that interrupt the manuscript.
- **Placeholder scan:** No TODO, TBD, FIXME, XXX, “insert here,” or placeholder markers were detected.
- **Result-value outliers:** $\sigma_8=0.794561$ and $S_8=0.813037$ are correctly identified as matched-control outputs. The alternate acoustic values are labeled parameter/transition trials rather than final results. $\Delta r_{\text{drag}}=2.457$ Mpc is a difference, not a mistaken ruler value.
- **Visual page integrity:** Low-resolution renders were produced for all 417 pages. No blank, clipped, rotated, or obviously corrupted pages were detected.

3. Exact wording corrections recommended

Current concept: “The complete gravitational action is ... with the high-density saturation response imposed through the modified background constraint.”

Replace with: “The covariant Einstein-plus-canonical-field action used for the matter and saturation sectors is The high-density saturated Friedmann constraint is introduced independently as an effective background modification; a covariant completion that derives this constraint remains open.”

Bounce clarification: “For $H=0$, differentiating the constraint and using total continuity gives the modified Raychaudhuri equation. Its value at $H=0$ is defined by the continuous limit of the regular evolution system. Equivalently, it is included as an independent effective evolution equation at the bounce.”

Stability clarification: “The positive kinetic matrix, unit scalar sound speeds, and positive potential Hessian establish linear stability of the retained canonical C-phi sector near the locked state. They do not by themselves establish stability of a future covariant completion of the high-density gravitational modification.”

Boltzmann clarification: “The post-bounce locked expansion history is embedded in CAMB and evolved through the full linear Einstein-Boltzmann hierarchy used in this effective implementation.”

4. Editorial cleanup list

- Remove spaces before commas and periods in displayed prose lists throughout the manuscript.
- Normalize spaces around symbols in tables: “1 .042322” -> “1.042322,” “13883 .67” -> “13883.67,” and similar extraction-visible formatting.
- Correct “Relativit” atstheorie” to “Relativitätstheorie” or an ASCII-safe bibliographic form if the production system cannot embed the glyph.
- Normalize journal abbreviations so punctuation is attached correctly: “Phys. Rev. D,” not “Phys. Rev. D ,”.
- Use one convention for “matched control,” “matched-control,” and “control model.”
- Use one convention for “drag horizon,” “sound horizon at drag,” and r_{drag} when referring to the same quantity.
- Add a version number and release date to the title or copyright page.
- Add page/bookmark navigation and a clickable table of contents.

5. Mathematical claim-boundary assessment

Internally demonstrated within the stated effective system: canonical field equations; field-to-radiation energy bookkeeping; recovery of the low-density Friedmann limit; a finite-density zero of H ; local stability of the potential minimum; the listed background and observational calculations, assuming the archived numerical implementation is exactly as described.

Not yet demonstrated by the PDF alone: a covariant action that yields the modified Friedmann law; uniqueness or microscopic derivation of the saturation correction; full gravitational perturbation stability; independent reproduction of CAMB modifications and likelihood calculations; a complete

primary-CMB likelihood with nuisance parameters; nonlinear structure formation; and observational preference over Lambda-CDM after parameter-complexity accounting.

6. Submission readiness decision

Decision: CONDITIONAL PASS FOR EXTERNAL TECHNICAL REVIEW. The manuscript is suitable to circulate to qualified researchers as an effective-model manuscript after the three high-severity claim/derivation clarifications are inserted and the visible punctuation artifacts are cleaned. It is not yet ready to claim independent reproducibility until the frozen code, CAMB modifications, parameter files, input data identifiers, and output tables are released.

Recommended release label: Version 1.0-RC1 (review candidate), not “verified.” After code release and correction of the items above, label the exact archived PDF and repository as Version 1.0.